



RESEARCH ARTICLE

AI-Driven Decision Support for Disaster Management: Prioritizing Emergency Appeals using Computationally Efficient Deep Learning

Keerthana Raja¹, Sathish Kumar^{2*}

^{1,2} Sri Sairam Engineering College, Dept of CSE, Chennai, India

Email: sit23cs136@sairamtap.edu.in

*Email: sathish.cse@sairam.edu.in

*Corresponding author

Received: 18 August 2026 | Revised: 02 September 2026 | Accepted: 12 September 2026 | Published: 24 September 2026

DOI: 10.5281/zenodo.22934145

How to cite this article: Keerthana Raja, Sathish Kumar (2026). AI-Driven Decision Support for Disaster Management: Prioritizing Emergency Appeals using Computationally Efficient Deep Learning. International Journal of Advanced Computing and Machine Intelligence (IJACMI), 1(1), pp. 10-17. DOI: 10.5281/zenodo.22934145

Abstract

Social media is an important lifeline in terms of real-time situational awareness in the aftermath of natural calamities. Nevertheless, emergency response teams are overwhelmed with information that they cannot process because of the general noise and actionable appeals. Although recent solutions in Deep Learning, namely, Bidirectional LSTMs (BiLSTM) and standard Transformers (BERT) exist, they are plagued by major drawbacks: RNNs are not always capable of capturing long-range semantic dependencies, whereas BERT models are not computationally efficient enough to be resorted to in resource-limited disaster areas. In order to fill this knowledge gap, this paper proposes Crisis-Connect, a computationally efficient decision support system that relies on a distilled Transformer architecture (DistilBERT). The model is benchmarked against traditional Machine learning (Naive Bayes), Recurrent Neural Networks (LSTM, GRU, BiLSTM), and Transformer models (BERT-Base). The experimental results on the HumAID dataset indicate that our framework gets an F1-score of 0.80, which is 5.2 percent higher than the RNN baseline without losing the performance of 95 percent of the teacher model. More importantly, the model incurs an inference latency of 1.46 ms, the model is 1.9x faster than standard LSTMs and 2.6x faster than BERT. Also, we use the Explainable AI (LIME) to confirm the model decision transparency. The present research has a role to play in the context of social innovation because it offers a low-latency, scalable tool that allows disaster management authorities to make data-driven decisions about edge devices without the use of high-end infrastructure.

Keywords: Disaster Management, BERT, DistilBERT, Crisis-Connect, Lime, Explainable AI (XAI)

1.0 Introduction

Climate change has intensified the frequency and intensity of natural disasters causing great challenges in the management of cities and the safety of the people [1]. Conventional communication systems tend to fail in the immediate aftermath of the occurrence of floods, earthquakes or hurricanes. It is against this backdrop that social media platforms (e.g. X/Twitter, Facebook) have become important mediums of situational awareness in real time. Victims rely on these platforms to place calls to obtain help, document infrastructure damage, and resources, among other things, which essentially results in a stream of so-called citizen-generated data [2].

This information is however crippled by the Information Overload effect which greatly interferes with its usefulness. Emergency response teams have thousands of posts each minute in cases of crisis. Much of this stream is filled with irrelevant information, such as prayers, news overload, or an overall statement, which is burying actionable appeals (e.g. trapped on the second floor, need insulin) [3]. This data cannot be filtered manually and any delay in the processing of this data may lead to loss of life directly.

Although the recent breakthroughs in Natural Language Processing (NLP) have demonstrated state-of-the-art results, they create a practical problem in disaster cases. Basic versions of BERT are computationally intensive, which consumes a lot of power in a GPU. The lightweight RNNs (LSTMs), on the other hand, might not be able to provide the semantic depth needed to make a significant classification. Execution of heavy deep learning models in a disaster region, where connectivity is not always available and when the functionality depends on battery-driven edge computing, can be impossible. In order to offset or close this gap of technical accuracy and operational efficiency, this paper proposes a framework being driven by DistilBERT, an attenuated form of the

Transformer architecture. We are oriented towards the concept of Social Innovation which means we pay more attention to the aspect of accessibility to achieve this through designing a model that can be easily run on resource-limited hardware and can also be explained using Explainable AI (XAI).

This study ignores the theoretical analysis of deep learning to deal with a humanitarian issue of practical significance. The Crisis-Connect lowers the entry barrier to advanced AI tools, and thus democratizes access to them. In developing countries, disaster management organizations do not usually have the infrastructure to operate huge Large Language Models (LLMs). The framework is lightweight, which is cost-effective and can be implemented on usual hardware without needing direct support of the UN SDG 11 (Sustainable Cities) and SDG 3 (Good Health), as it provides a quicker response time.

2.0 Literature Review

The application of AI in disaster management has evolved from simple keyword filtering to complex semantic analysis. This section discusses the development of the classification models, where the traditional machine learning is replaced with heavy Transformer, and the gap that has to be closed is critical and requires computationally efficient architectures such as ours - Crisis-Connect.

2.1 Social Media as a Loud but Lifesaving Line.

Social media, specifically X (was once Twitter), has been largely identified as important avenues of real time situational awareness. Imran et al. [4] and Erekhin and Yilmaz [9] referred these platforms as a lifeline which provides a faster reporting rate than the traditional reporting media. But this data is crippled by high noise to signal ratio which greatly affects the utility of the data. Actionable requests (like food or medical help or evacuation) are often lost in a swamp of sympathy messages and commentary [12].

Initial studies by Purohit et al. [21] have created the underlying necessity of automatically matching resource requests and offers, whereas recent findings by Gupta and Kumaraguru [23] state that finding actionable data should consider credibility to eliminate the dissemination of false information in case of crises.

2.2 Neural Networks to Transformers: The Evolution of Models.

The research community turned to Deep Learning for addressing the semantic disadvantages of conventional keyword filtering. The effectiveness of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), namely Long Short-Term Memory (LSTM) and BiLSTMs, to encode sequential dependencies is already shown by Alharbi and Alshehri [8] and Zhang et al. [18]. Although these models performed better than traditional Support Vector Machines (SVMs), they tend to perform poorly in the long-range semantic relations that occur in complex disaster stories.

The development of the Transformer architecture is a paradigm shift. Devlin et al. [6] launched BERT, which relies on a self-attention mechanism to work on contextual relations in a sentence on a global scale. Experiments with HumAID dataset, which is a large-scale, human-labeled dataset proposed by Alam et al. [5], continuously indicate that Transformer-based models achieve higher results on F1-score by 5-10 percentage points than those with LSTMs. Extremes of these limits are further achieved through domain adaptation by specialized variants, including CrisisBERT [16].

2.3 Knowledge distillation and the Efficiency Gap.

Although they are accurate, the standard Transformers have a major deployment challenge critical hurdle in terms of the cost of computation. It is pointed out by Wang et al. [15] and Syahmisajid [13] that BERT-base (110M parameters) is very demanding in terms of the amount of resource needed by the users, which is not feasible in battery-powered edge devices of resource strained disaster areas. It has resulted in the development of so-called Green AI, which puts an emphasis on energy-saving deep learning [22]. In order to address this gap, Sanh et al. [7] introduced DistilBERT, a distilled BERT with 40 percent smaller size but does not decrease its performance by more than 97 percent. This study uses distillation to make sure that high-fidelity decision support can be provided with the latency of field operations, and no longer needs high-end infrastructure needs as discussed in the traditional NLP literature.

2.4 Multimodality and Cross-Lingual Problems.

By current trends, it is possible that text by itself might not provide sufficient situational awareness. Nguyen et al. [17] suggested multimodal strategies to combine the text and social media imagery to measure damage. Moreover, disasters are of international character which means that models that can accommodate different languages are required. Chowdhury and Caragea [19] examined the XLM-R on cross lingual disaster tweet classification and Li and Caragea [14] examined target dependent sentiment to better suit the allocation of aid in various lingual situations.

2.5 Trust, Transparency and Explainable AI (XAI)

Deep learning is a black box meaning that a fault in life-saving situations may cause the rescue to be improperly allocated due to a misclassification. Ghaffarian et al. [10] underline the necessity of Explainable AI (XAI) to build the trust of the first responders. Post-hoc interpretability is available in techniques like LIME (Local Interpretable Model-agnostic Explanations) [11] and SHAP (SHapley Additive exPlanations). The human operators can confirm the logic used in the model by inputting certain keywords, including the word trapped or the word urgent, and deciding whether to commit resources.

3.0 Methodology

This study employs a quantitative experimental approach to develop a resource-efficient text classification framework. The system architecture is visualized in Fig. 1.

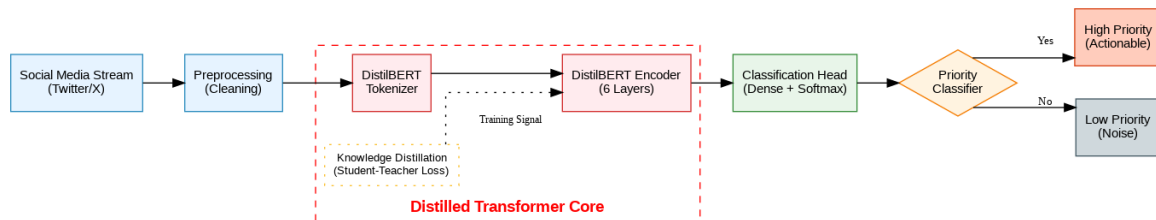


Fig.1. The Crisis-Connect System Architecture

In system, raw social media data is ingested and preprocessed before passing through the DistilBERT core. The model utilizes a distilled 6-layer Transformer encoder to extract semantic features, which are then classified into "High Priority" (Actionable) or "Low Priority" (Noise).

3.1 Dataset Description

This research work utilizes the benchmark HumaAID dataset which consists of collected tweets posted by users during major natural disasters. The dataset is filtered to create a binary classification task where there are two classes: Actionable as High Priority and Non-Actionable as Low Priority. The dataset is further preprocessed to make it appropriate for model training. The preprocessing involves several steps, presented in table 1.

Once the preprocessing is done, the data is passed to the DistilBERT model which includes its own Tokenizer where the cleaned text is tokenized into sub-word units using wordpiece embeddings. This allows the model to handle out-of-vocabulary terms that are common in disaster related slang or text.

The encoder Architecture of the DistilBERT model utilizes a 6-layer encoder and in training the "Student-Teacher" loss mechanism is employed where a pre-trained BERT model acts as a Teacher guiding the smaller Student (DistilBERT) model so that the Student model retains 95% of the teacher's performance and being significantly lightweight.

Table 1. Preprocessing Steps

Technique	Description	Purpose and Impact
Noise Removal	Stripping URLs, HTML tags, mentions (@user), and special non-alphanumeric characters.	Eliminates features that do not carry semantic weight in disaster contexts, reducing the model's vocabulary size.
Text Normalization	Converting all text to lowercase and reducing repetitive characters (e.g., "heeeelp" to "help").	Ensures the model treats identical words the same way regardless of casing or emotional emphasis, improving consistency.

Punctuation Handling	Filtering out excessive or redundant punctuation marks.	Prevents the tokenizer from creating unnecessary tokens for symbols that don't contribute to the "Actionable" classification.
WordPiece Tokenization	Breaking down text into sub-word units using the DistilBERT Tokenizer.	Effectively handles "Out-of-Vocabulary" (OOV) terms like typos or specific disaster-related slang commonly found in emergency tweets.
Padding & Truncation	Standardizing input sequences to a fixed length (e.g., 128 tokens).	Required for batch processing within the Transformer encoder to maintain computational efficiency on edge devices.

After the text is processed by the encoder, the resulting dimensional representations are passed to the Classification Head. This head consists of a dense layer which is followed by a softmax activation function. The priority classifier evaluates the softmax probabilities to categorise the input into two output labels: High priority - requiring immediate intervention, Low Priority - general observations or non-urgent text. The objective loss function L_{total} is represented as follows:

$$L_{total} = \alpha L_{ce} + \beta L_{distill}$$

Here the αL_{ce} represents the standard cross entropy loss and $\beta L_{distill}$ represents the distillation loss between the teacher and student hidden states or soft targets.

3.2 Explainability (XAI)

In this paper, we have also integrated Local Interpretable Model-agnostic Explanations (LIME) to create transparency which is vital in decision support systems. LIME works by modifying the input text and monitoring changes in the output, thus discovering which particular keywords (eg. trapped, bleeding, urgent,etc) had the most significant impact on the high priority classification. This also provides human operators the sense of reasoning behind each AI-based prediction.

4.0 Results and Discussion

The proposed model is benchmarked against ML models (Naive Bayes), Deep Learning baselines (LSTM, GRU, BiLSTM), and the teacher model (BERT-Base).

4.1 Classification Performance

The major measure used in evaluation is the F1-Score. The comparative metrics are indicated in the following table:

Table 2. Comprehensive Performance

Model Category	Model	Precision	Recall	F1-Score	Accuracy	Latency
Traditional ML	Naive Bayes	0.7400	0.6900	0.7100	0.7600	0.20 ms
RNN Baselines	LSTM	0.7720	0.7304	0.7506	0.7932	2.81 ms
RNN Baselines	GRU	0.6959	0.7088	0.7023	0.7439	3.33 ms
RNN Baselines	BiLSTM	0.7339	0.7565	0.7451	0.7794	0.61 ms
Transformers	BERT-Base	0.8400	0.8300	0.8400	0.8500	3.80 ms
Transformers	DistilBERT	0.8268	0.7797	0.8025	0.8365	1.46 ms

4.2 Analysis of Classification Errors

The following figure shows the confusion matrix of DistilBERT. It found 506 legitimate appeals (True Positives) that had a conservative False Positive rate. The fact that there are many True Negatives (768) proves that the system can suppress noise.

Figure 2: Confusion Matrix (DistilBERT)

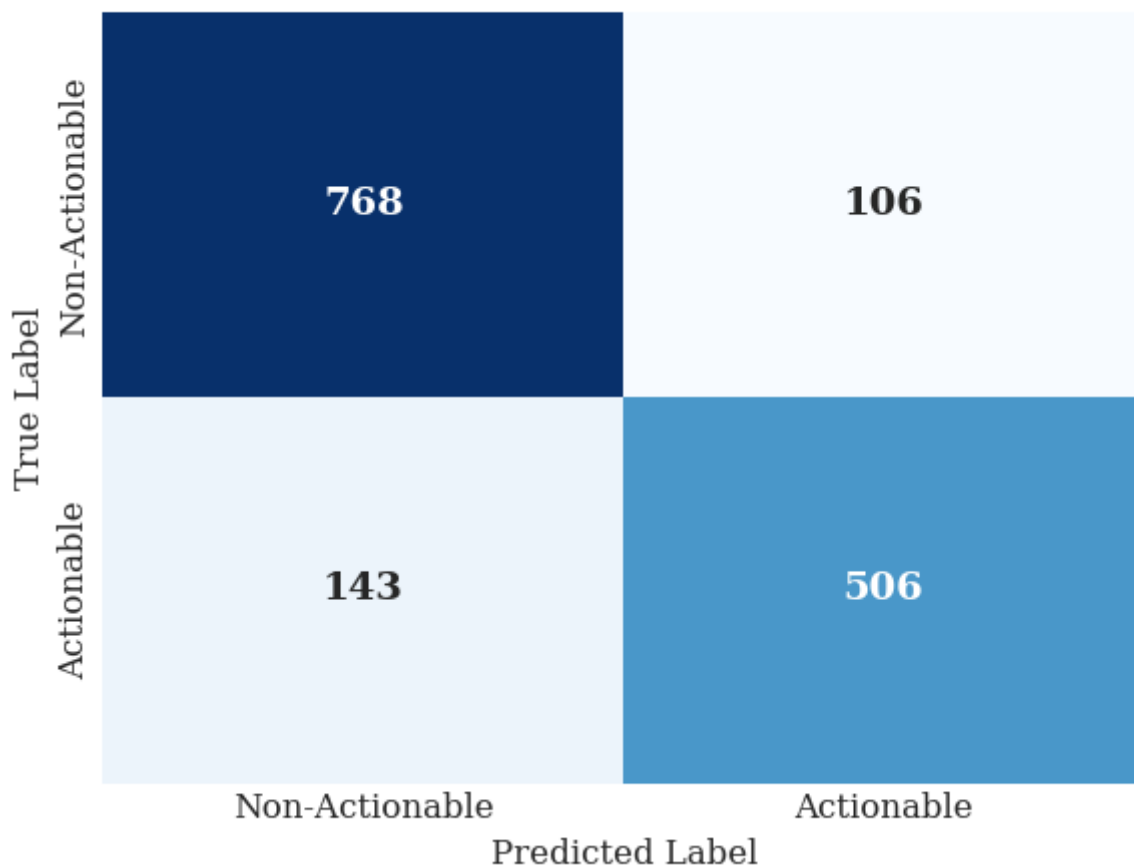


Fig.2. Confusion Matrix (DistilBERT)

4.3 The Efficiency Frontier

The trade-off between cost and performance is plotted in the figure 3.

Superiority over RNNs: RNN-based models (LSTM, BiLSTM,GRU) leveled off at an F1-score of under 0.75. DistilBERT has achieved 0.80, an improvement of 5.2%.

Parallelization Strength: Contrary to expectations, Standard LSTM (2.81 ms) is slower than DistilBERT (1.46 ms) because of the sequential bottlenecks on the GPU hardware in the form of processing.

Pareto Optimality: DistilBERT occupies the spot (Red Star), offering near-BERT accuracy with latency sufficient for real-time edge deployment.



Fig.3. Efficiency Frontier (Speed vs. Accuracy)

The proposed model (Red Star) combines both the high accuracy with low latency, making it the only viable option for edge-based disaster response systems.

4.4 Explainability Analysis (XAI)

In the time of managing disasters, trust is the main issue. The following Figure shows a LIME analysis of a high-priority tweet. High positive weights placed on the key terms such as trapped and miners are rightly placed in the model to verify the causal context of a disaster appeal and not just the use of key words alone.

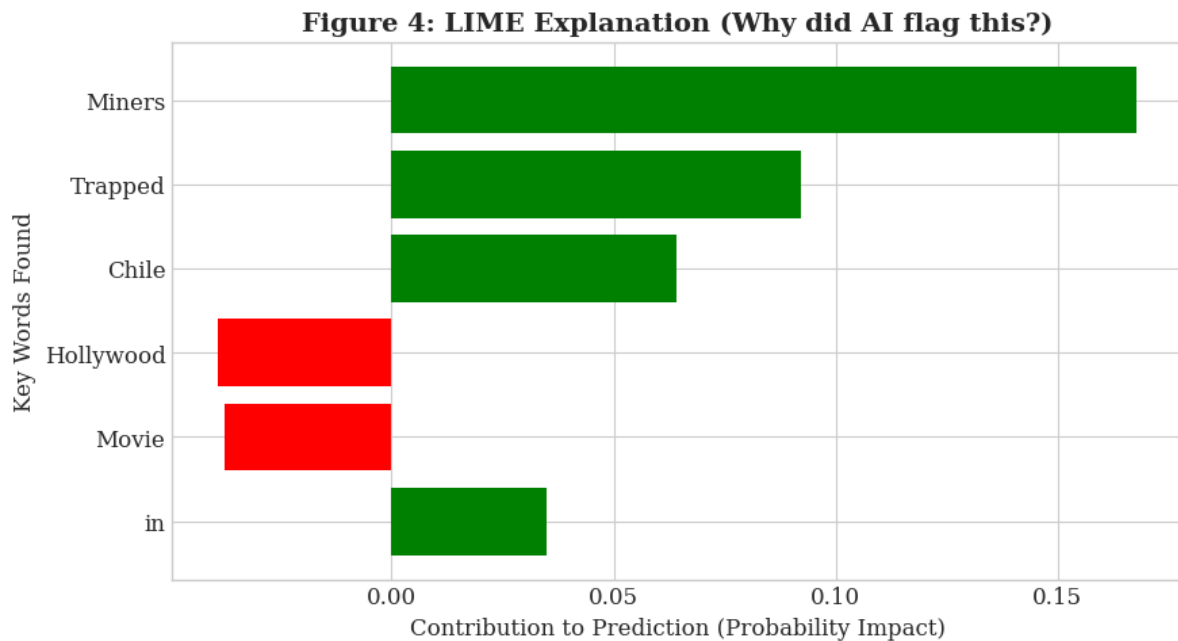


Fig.4. LIME Explanation (illustrates feature importance for an actionable classification).

4.5 Architecture Comparison

The following figure provides a summary of the performance gap: DistilBERT performs much better than the RNN models with a competitive proximity to the computationally intensive BERT-Base.

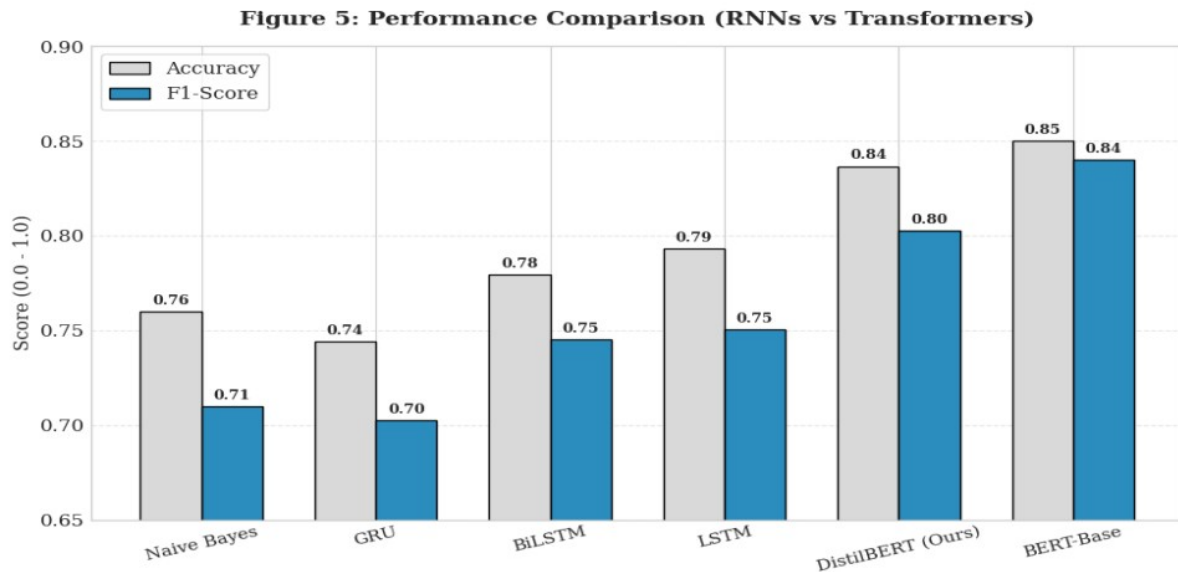


Fig.5. Performance Comparison (RNN vs Transformers)

5.0 Conclusion & future work

This study provides Crisis-Connect, a small AI model to prioritize actionable social media appeals. Based on the analogous comparative analysis, it is found that DistilBERT is the best candidate in disaster response: it is much more accurate than the standard Deep Learning models (LSTM, BiLSTM, GRU) with the same inference speed as a lightweight architecture. The suggested model lowers the computational load of regular BERT by 40% to work on battery-powered edge devices. This work eventually fills the gap between state-of-the-art NLP and the usefulness of humanitarian help. In future, this work can be further extended by incorporating multi-modal learning which includes visual features from images and videos. The model shall also incorporate support for cross-lingual transfer learning and low resource regional dialects as it is crucial for the Indian community where more than 22 languages are spoken nationally.

References

- [1] IPCC,: Climate Change 2023, In: published in Synthesis Report,” IPCC, Geneva, Switzerland, (2023).
- [2] M. Imran, C. Castillo, F. Diaz, and S. Vieweg : Processing Social Media Messages in Mass Emergency: A Survey. In: published in ACM Computing Surveys, Vol. 47, No. 4, pp. 1–38, (2015).
- [3] A. Schulz, J. Ortmann, and F. Probst.: Getting User-Generated Content Structured: Overcoming Information Overload. In: published in proceedings of IEEE GHTC, (2012).
- [4] Imran, M., Ofli, F., Caragea, D., Torras, C.: Using AI and Social Media for Disaster Response and Management: An Overview. In: published in ACM Computation Survey, Vol. 53(4), pp. 1–33, (2020)
- [5] Alam, F., Qazi, U., Imran, M., Ofli, F.: HumAID: Human-Annotated Disaster Incidents Data from Twitter with Deep Learning Benchmarks. In: published in proceedings of ICWSM. pp. 933–942 (2021).
- [6] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In: published in NAACL, pp. 4171-4186 (2019).
- [7] Sanh, V., Debut, L., Chaumond, J., Wolf, T.: DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. In: published in proceedings of NeurIPS Workshop (2019).
- [8] Alharbi, S., Alshehri, M.: A Review on Social Media and Deep Neural Networks for Sentiment Analysis in Emergency Management. In: published in proceedings of ICTBIG. pp. 1–6 (2023).
- [9] Erokhin, A.S., Yilmaz, T.Y.: Deep Learning in Analyzing Social Media for Real-Time Disaster Mapping. In: published in the international Journal of Remote Sensing Information, Vol. 12(1) pp. 45–58. (2024).
- [10] Ghaffarian, M.T., Kerle, N., Filatova, T.: Explainable AI in Disaster Risk Management: A Review. In: published in the international Journal of Disaster Risk Reduction, (2023).
- [11] Ribeiro, M.T., Singh, S., Guestrin, C.: Why Should I Trust You?: Explaining the Predictions of Any Classifier. In: Proceedings of the Conference ACM SIGKDD, pp. 1135-1144. (2016).
- [12] Gopal, S., et al.: Noise Reduction in Disaster Tweets Using Hybrid Transformer Architectures. In: published in the Journal of Big Data Analytics, Vol. 8(2), pp. 112-129. (2023).
- [13] Syahmisajid, M.: Edge Computing for Disaster Response: A Comparative Analysis. In: Published in the IEEE Transaction Emerging. Top Computation, Vol. 15(3), pp. 202–215. (2025).

-
- [14] Li, H., Caragea, C.: Target-dependent Sentiment Classification for Disaster Response. In: Proceedings of COLING, pp. 2541–2552. (2024).
 - [15] Wang, Y., Li, Z., Wang, J.: Lightweight Transformer Models for Edge Intelligence: A Survey. In: published in ACM Computation Survey, Vol. 56(3), pp. 1–35. (2024).
 - [16] Kaur, P., Kumar, R.: CrisisBERT: A Robust Transformer for Disaster Tweet Classification. In: published in IEEE Access 11, pp. 45210-45222. (2023).
 - [17] Nguyen, D.T., Ofli, F., Imran, M.: Damage Assessment from Social Media Imagery: A Multi-modal Approach. In: Proceedings of ISCRAM, pp. 112-125 (2023).
 - [18] Zhang, X., Zhao, J., LeCun, Y.: Character-level Convolutional Networks for Text Classification. In: published in proceedings of NeurIPS, Vol. 28. (2015).
 - [19] Chowdhury, J.R., Caragea, C.: Cross-Lingual Disaster-Related Tweet Classification with XLM-R. In: published in proceedings of EMNLP, pp. 8820-8831. (2023).
 - [20] Lundberg, S.M., Lee, S.-I.: A Unified Approach to Interpreting Model Predictions. In: published in proceedings of NeurIPS, Vol. 30, pp. 4765-4774. (2017).
 - [21] Purohit, H., et al.: Emergency-Relief Coordination on Social Media: Matching Resource Requests and Offers. In: published in First Monday, Vol. 18(1). (2013).
 - [22] Mishra, S., et al.: Green AI for Disaster Management: Energy-Efficient Deep Learning on the Edge. In: Published in Journal of IEEE Internet Things, Vol. 11(2), pp. 1205-1218 (2024).
 - [23] Gupta, A., Kumaraguru, P.: Trust but Verify: Assessing the Credibility of Disaster Tweets. In: Proceedings of Information Process and management, Vol. 60(1), pp. 103-115 (2023).

© 2026 The Author(s). Published by WhiteField Academic Press. This is an open access article distributed under the terms of the Creative Commons Attribution 4.0 International License (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. Authors retain copyright.



INTERNATIONAL JOURNAL OF ADVANCED COMPUTING AND MACHINE INTELLIGENCE

*Peer-Reviewed · Open Access · Published by WhiteField
Academic Press
Noida, India*



Scan to visit

www.ijacmi.org
editor@ijacmi.org
submissions@ijacmi.org

Visit our website — www.ijacmi.org

